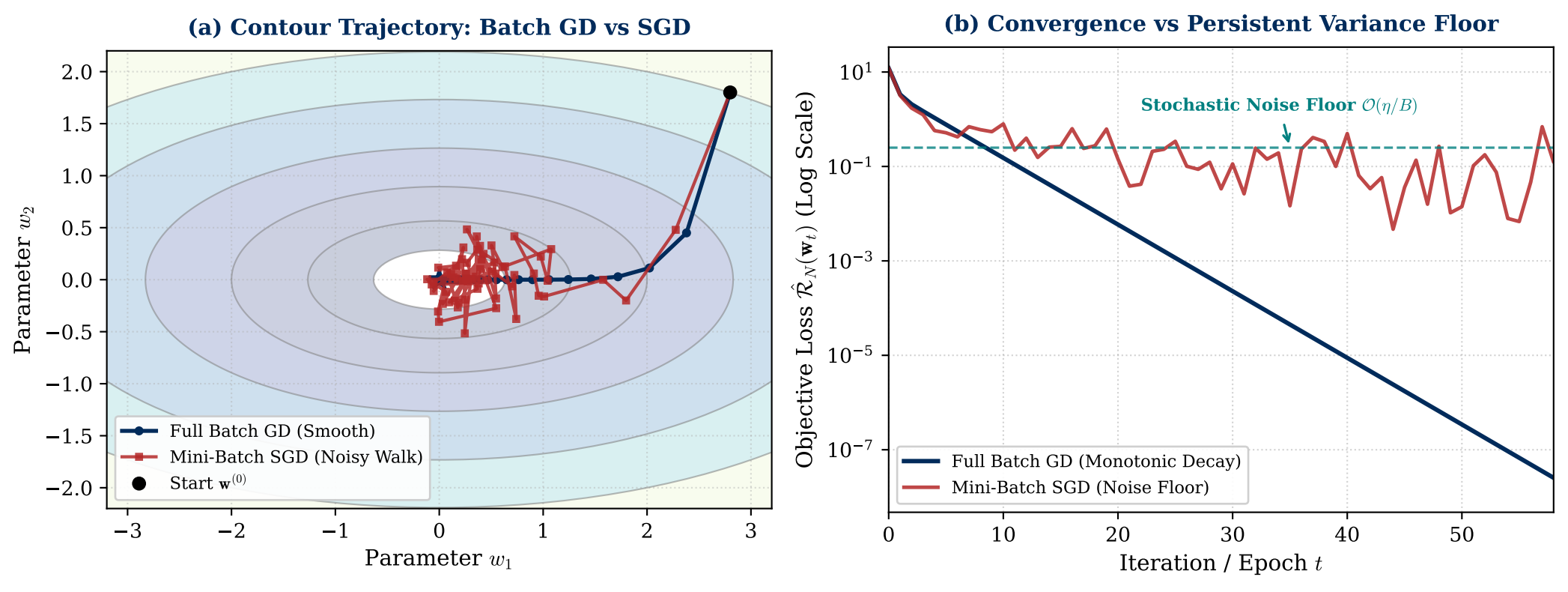

(a) Contour Trajectory: Batch GD vs SGD
Parameter w_2
Parameter w_1
Full Batch GD (Smooth)
Mini-Batch SGD (Noisy Walk)
Start $\mathbf{w}^{(0)}$
(b) Convergence vs Persistent Variance Floor
Objective Loss $\hat{\mathcal{R}}_N(\mathbf{w}_t)$ (Log Scale)
Stochastic Noise Floor $\mathcal{O}(\eta/B)$
Full Batch GD (Monotonic Decay)
Mini-Batch SGD (Noise Floor)
Iteration / Epoch t